George Nicolas Diaz-Wahl

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Research Interests: Arithmetic geometry, p-adic Hodge theory, prismatic cohomology, Galois representations.

Education

Purdue University Aug 2021 – Present
Ph.D. in Mathematics, expected May 2027 *West Lafayette, IN*
Advisor: Tong Liu

Carnegie Mellon University Aug 2017 – Mar 2021
B.S./M.S. in Mathematics; thesis: The rank of modular Hecke algebras *Pittsburgh, PA*
Advisor: Carl Wang-Erickson

Research

Papers and Work in Progress

Semistable $\mathbb{Z}_p$-local systems and Breuil modules
Preliminary draft; preprint forthcoming.

A relative integral comparison of p-adic étale and log crystalline cohomology
In preparation.

Prismatic F-gauges and derived Breuil–Kisin modules
In preparation.

Selected Research Experience

IU–Purdue Joint Workshop on Prismatic F-gauges Apr 2026
Purdue University in Indianapolis *Indianapolis, IN*

- Presented work in progress on prismatization and syntomification over imperfect fields.

Arizona Winter School 2024: Abelian Varieties Mar 2024
University of Arizona *Tucson, AZ*

- Rachel Pries' project group on p-rank of double covers of elliptic curves.
- Studied the dimension of moduli spaces of double covers of elliptic curves with fixed p-rank and existence questions for supersingular singular double covers.

Honors Master's Thesis Research Mar 2020 – Mar 2021
Carnegie Mellon University *Pittsburgh, PA*

- Expository thesis, advised by Carl Wang-Erickson, relating congruences of modular forms to ranks of p-adic Hecke algebras.

Oregon State University REU Jun 2019 – Aug 2019
Oregon State University *Corvallis, OR*

- Studied (pre)periodic points of power series over Henselian fields; results appeared in the 2019 Oregon State REU Proceedings.

Selected Seminar Talks

Prismatization for imperfect fields, IU–Purdue Joint Workshop on Prismatic F-gauges — April 2026
Prismatic cohomology, Learning Seminar on Prismatic F-Gauges — Spring 2026
Étale sheaves and cohomology, Learning Seminar on the Weil Conjectures — Fall 2024
p-rank stratification in the moduli space of double covers of elliptic curves, Arizona Winter School — Spring 2024
The Taylor–Wiles method and patching, Learning Seminar on Modularity Lifting — Fall 2023
Galois representations and (φ, Γ)-modules, Learning Seminar on the Emerton–Gee Stack — Fall 2022

Teaching Experience

Purdue University Aug 2021 – Present
West Lafayette, IN

- **Grader** (Spring 2022–present): MA 511, Linear Algebra with Applications (2 semesters); MA 453, Abstract Algebra (2 semesters); MA 553, Graduate Algebra (3 semesters); MA 454, Honors Abstract Algebra; MA 590.
- **Teaching Assistant**: MA 261, Calculus III (Fall 2025).
- **Instructor**: MA 108, Mathematics as a Profession (Fall 2024).
- **Teaching Assistant**: MA 162, Calculus II (Fall 2021).

Summer Math Teacher Summers 2022, 2024, 2025, 2026
Harker Upper School *San Jose, CA*

- Jump-start to Algebra II (Summers 2025, 2026).
- Algebra I Honors (Summer 2024).
- Algebra II/Trigonometry Honors (Summer 2022).

Summer Math Teacher Aug 2024; Jul–Aug 2025
Art of Problem Solving Academy *Mountain View/San Jose, CA*

- High School Math Contests and Middle School Math Contests (July–August 2025).
- Math Level 4 (August 2024).

Mathematics Instructor Jul–Aug 2023; Dec 2024
AlphaStar Academy *San Jose, CA*

- AMC 10 and AMC 12 courses (December 2024; July–August 2023).

Undergraduate Teaching Assistant & Grader Aug 2018 – Mar 2021
Carnegie Mellon University *Pittsburgh, PA*

- Concepts of Mathematics (2 semesters) and Algebraic Structures (3 semesters).

Seminar Organization

Organized ∞-Categories and Higher Algebra learning seminar	Spring 2026
Organized p-adic Geometry learning seminar	Fall 2025
Organized Arithmetic student seminar	Spring 2023, Fall 2024
Organized Moduli Spaces and Algebraic Stacks seminar	Fall 2023
Coorganized Commutative Algebra student seminar	Spring 2023

Honors & Awards

Excellence Award in Mathematics
Purdue University *Apr 2026*

Special Award
Purdue University *Mar 2024*
Award of $1500 for excellence in teaching and organization of the Moduli Spaces and Algebraic Stacks seminar.

Ross Fellowship
Purdue University *Aug 2021 – May 2025*